

AhamX Bodhi

Where I Becomes Infinite

One-Shot Prompting

Module 2.3: Prompt Engineering | Prof. Sashikumaar Ganesan |
IISc Bengaluru



Session Overview

What You Will Learn

- What one-shot prompting is
- How a single example guides LLM behaviour
- When to use one-shot vs. zero-shot
- How to choose an effective example

Session Details

- **Duration:** 7 minutes
- **Bloom's Level:** B2 — Understand
- **Difficulty:** L1 — Beginner
- **Module:** 2.3 Prompt Engineering
- **Prereq:** Zero-Shot Prompting



Learning Objectives

By the End of This Session You Will Be Able To

- **Define** one-shot prompting and its role in in-context learning
- **Explain** how a single example shapes model output
- **Construct** effective one-shot prompts for format-sensitive tasks
- **Compare** one-shot and zero-shot approaches for given scenarios





One-Shot Prompting: Definition

Core Definition

One-shot prompting provides the model with *exactly one* input-output example before presenting the target task. The single example communicates the desired format, style, and scope of the expected response.

Contrast with Zero-Shot

- Zero-shot: Instruction + Input only
- One-shot: Instruction + **One Example (Input + Output)** + New Input
- The example acts as a demonstration, not additional training
- No model weights change; this is *in-context learning*



In-Context Learning: The Mechanism

How the Model Uses the Example

- The model processes the entire prompt (instruction + example + new input) as one sequence
- It identifies the pattern from the example and applies it to the new input
- No gradient updates occur; the model leverages its attention mechanism
- This is called **in-context learning (ICL)**: learning from context, not training

Key Distinction

One-shot learning in prompting is **not** the same as one-shot learning in meta-learning. Here it refers to providing one demonstration in the prompt context, not training on one example.



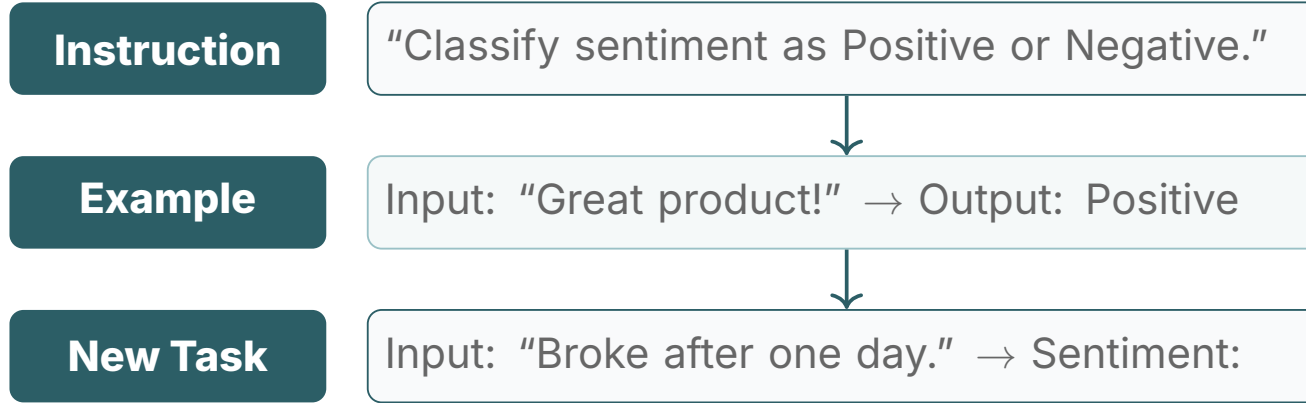
Anatomy of a One-Shot Prompt

Structure

- **Instruction:** "Classify the sentiment of customer reviews as Positive or Negative."
- **Example Input:** "Review: 'The delivery was fast and the product works perfectly.' "
- **Example Output:** "Sentiment: Positive"
- **Separator:** "—" or a blank line to distinguish example from new task
- **New Input:** "Review: 'The screen cracked after two days of use.' "
- **Output Prompt:** "Sentiment:" (leave blank for model to complete)



One-Shot Prompt: Visual Structure



The model completes "Sentiment:" by applying the pattern learned from the example.



Choosing an Effective One-Shot Example

Principles for Example Selection

- **Representative:** Example should reflect the typical case, not an edge case
- **Unambiguous:** The example input-output pair should have only one clear interpretation
- **Format-demonstrating:** If output format matters, show it exactly in the example
- **Same domain:** Example and target input should come from the same distribution
- **Correct:** A wrong example actively misleads the model



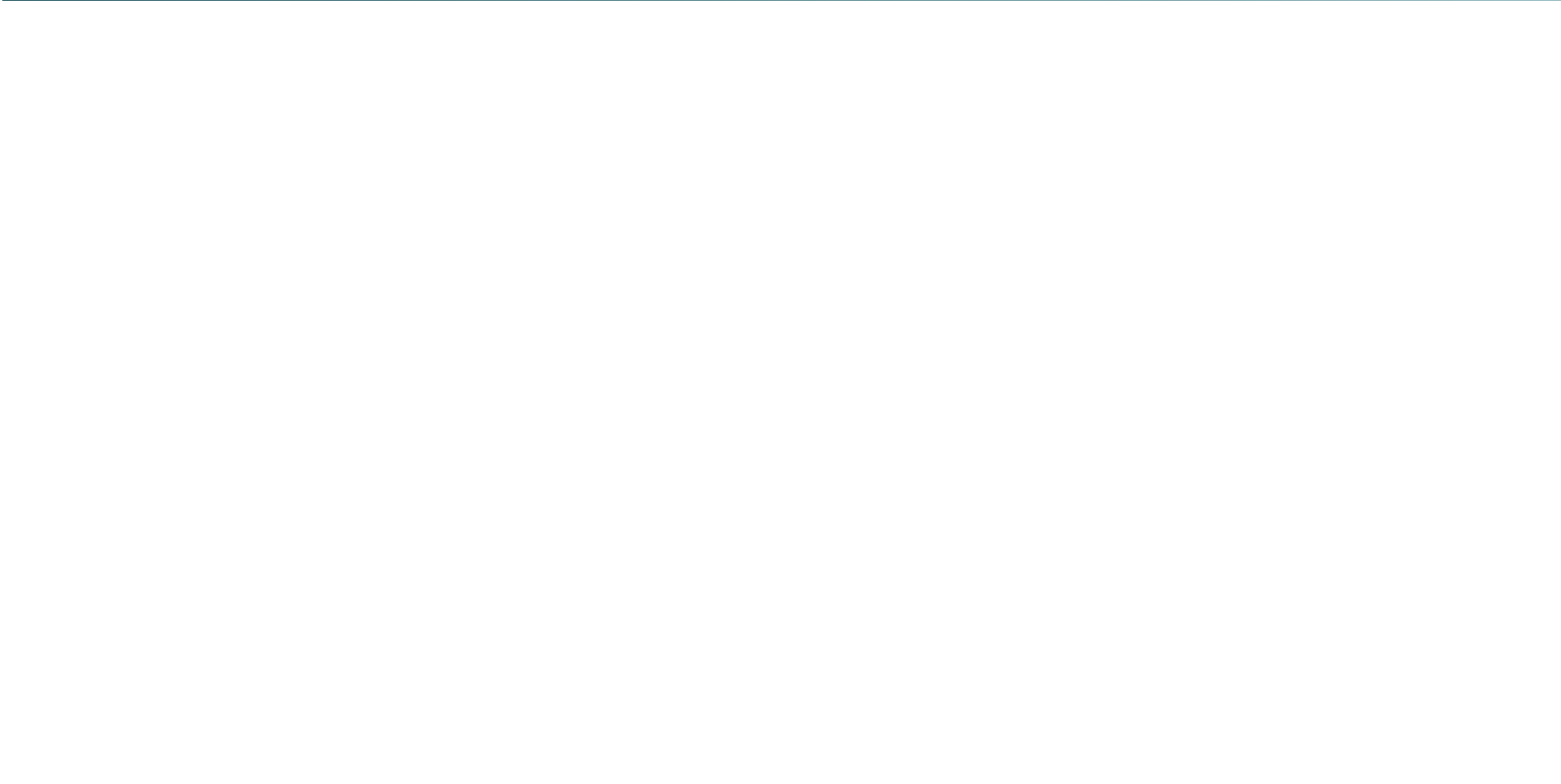
When One-Shot Outperforms Zero-Shot

Scenarios Where One Example Helps

- **Custom output formats:** Proprietary JSON schemas, specific markup languages
- **Style matching:** Formal legal language, brand tone, technical style guides
- **Novel task types:** Tasks the model may not have encountered frequently
- **Ambiguous instructions:** One example resolves ambiguity faster than lengthy description

Rule of Thumb

- Start with zero-shot; upgrade to one-shot if output quality is inconsistent
- One-shot costs approximately 2x the tokens of zero-shot for the same task





Application: Structured Data Extraction

Invoice and Form Processing

Companies processing invoices, receipts, and forms use one-shot prompts to extract structured fields into JSON or database records. One example shows the model the exact schema expected, eliminating the need for custom extraction models or labeled datasets.

One-Shot Extraction Pattern

- Show one annotated invoice → extracted JSON pair
- Model applies the same schema to new invoices
- Handles vendor-specific variations through format generalization
- Deployed by companies such as Rossum, Hypatos, and Infrd



Application: Branded Email and Content Generation

Maintaining Tone and Style

Marketing automation platforms (HubSpot, Mailchimp AI, Jasper) use one-shot prompts to ensure generated emails match a brand's established voice. One exemplary email establishes the tone benchmark for all subsequent generations.

Style Anchoring with One Shot

- Instruction: "Write a promotional email in the same style as the example."
- Example: One approved brand email (input brief → output email)
- New task: New campaign brief
- Result: Consistent brand voice without extensive prompt engineering



Application: Natural Language to SQL

Database Query Automation

BI tools (Tableau Pulse, Looker AI, ThoughtSpot) use one-shot prompts to convert natural language questions into SQL. One example teaches the model the specific database schema naming conventions.

One-Shot NL2SQL

- Example: "How many users signed up last month?" → `SELECT COUNT(*) FROM users WHERE . . .`
- New query: "What is the total revenue by region for Q3?"
- The model applies schema and syntax conventions from the example



Zero-Shot vs. One-Shot: Comparison

Dimension	Zero-Shot	One-Shot
Examples provided	0	1
Token cost	Lowest	Low
Format control	Instruction-based	Example-demonstrated
Best use case	Common NLP tasks	Custom formats, style matching
Failure mode	Misunderstood format	Wrong example misleads model





Key Takeaways

Core Concepts to Remember

- One-shot prompting provides exactly one input-output example in the prompt
- It leverages in-context learning: no model weights are updated
- A single example communicates format, style, and scope more powerfully than description
- Example quality is critical: a wrong example actively hurts performance
- Use one-shot when zero-shot produces inconsistent formats or style mismatches

AhamX Bodhi

Where I Becomes Infinite